

Security Groups and NACL

Class no:18

Security Groups (SG)

- **Scope:** Operates at the **instance level** (attached to EC2 instances).
- **Rule type:** Only **allow rules** (no explicit deny).
- **Evaluation:** All rules are evaluated together.
- **Stateful** – return traffic is automatically allowed if the request was allowed.
- **Default behaviour:** Denies all inbound traffic, allows all outbound traffic.
- **Use case:** Best for controlling access to specific EC2 instances (like allowing SSH, HTTP, HTTPS).

Network Access Control Lists (NACLs)

- **Scope:** Operates at the **subnet level** (applies to all resources in the subnet).
- **Rule type:** Supports both **allow and deny rules**.
- **Evaluation:** Rules are evaluated in order, starting from the lowest rule number.
- **Stateless** – return traffic must be explicitly allowed.
- **Default behaviour:** Allows all inbound and outbound traffic unless rules are added.
- **Use case:** Best for broader subnet-level control, like blocking specific IP ranges or enforcing stricter rules across multiple instances.

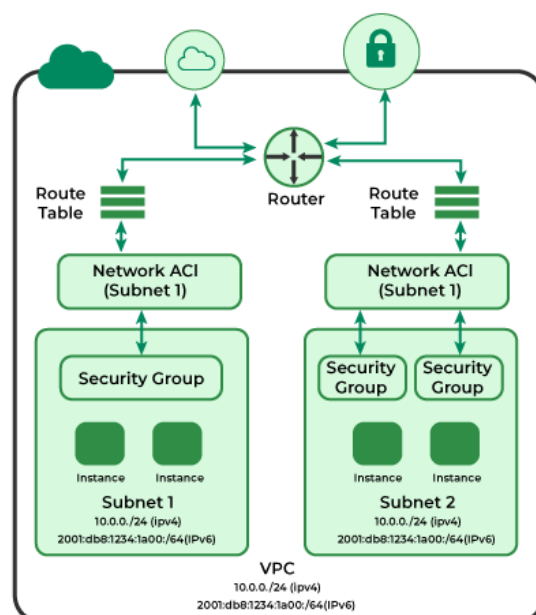


Fig. Architecture of Security group and NACL

Differences Between SG and NACL

Feature	Security Group (SG)	Network ACL (NACL)
Level	Instance (EC2)	Subnet
Rules	Allow only	Allow + Deny
Evaluation	All rules together	Rule order matters
Statefulness	Stateful	Stateless
Return traffic	Auto allowed	Must be allowed
Default inbound	Deny all	Allow all
Default outbound	Allow all	Allow all
Priority	No priority concept	Lowest rule wins

Hands on: Creating security group and NACL

Step no:01 Creating security group.

VPC → security group → Create Security Group.

Name: My-Security-Group

VPC: Choose the VPC

By default,

Inbound rules → none

Outbound rules → All traffic

(Note: Security groups has **only allow** option, **no deny** option.)

Basic details [Info](#)
Security group name [Info](#)

Name cannot be edited after creation.
Description [Info](#)

VPC [Info](#)

Inbound rules [Info](#)
This security group has no inbound rules.
[Add rule](#)

Outbound rules [Info](#)
Type [Info](#) Protocol [Info](#) Port range [Info](#) Destination [Info](#) Description - optional [Info](#)

All traffic

All

All

Custom

Delete

[Add rule](#)

Rules with destination of 0.0.0.0/0 or ::/0 allow your instances to send traffic to any IPv4 or IPv6 address. We recommend setting security group rules to be more restrictive and to only allow traffic to specific known IP addresses.

Tags - optional
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.
No tags associated with the resource.
[Add new tag](#)
You can add up to 50 more tags

[Cancel](#) [Create security group](#)

Step no:02 Create a NACL.

VPC → Security → NACL → Create NACL.

Name: my-nacl-01

VPC: our VPC Id

Create network ACL [Info](#)

A network ACL is an optional layer of security that acts as a firewall for controlling traffic in and out of a subnet.

Network ACL settings

Name - optional

Creates a tag with a key of 'Name' and a value that you specify.

my-nacl-01

VPC

VPC to use for this network ACL.

vpc-043c7b3268ebf3f1c

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Q Name



Value - optional

Q my-nacl-01



Remove tag

Add tag

You can add 49 more tags

Cancel

Create network ACL

By default,

Inbound rules → All traffic

Outbound rules → All traffic

acl-0ee38e03681596a32 / my-nacl-01

Details [Info](#)

Network ACL ID

acl-0ee38e03681596a32

Associated with

-

Default

No

VPC ID

vpc-043c7b3268ebf3f1c

Owner

314383175192

Inbound rules

Outbound rules

Subnet associations

Tags

Inbound rules (1)

Q Filter inbound rules

Rule number	Type	Protocol	Port range	Source	Allow/Deny
*	All traffic	All	All	0.0.0.0/0	Deny

acl-0ee38e03681596a32 / my-nacl-01

Actions

Details [Info](#)

Network ACL ID

acl-0ee38e03681596a32

Associated with

-

Default

No

VPC ID

vpc-043c7b3268ebf3f1c

Owner

314383175192

Inbound rules

Outbound rules

Subnet associations

Tags

Outbound rules (1)

Q Filter outbound rules

Rule number	Type	Protocol	Port range	Destination	Allow/Deny
*	All traffic	All	All	0.0.0.0/0	Deny

We can add inbound rules and outbound rules. They execute or evaluate from the lower number rule first.

For ex. If we have two inbound rules,

99 → All traffic → deny

101 → All traffic → allow

As **99** is the lowest rule number, it will deny all traffic, even though we allow it in rule number **101**.

Inbound rules

Outbound rules

Subnet associations

Tags

Inbound rules (3)

Filter inbound rules

Rule number	Type	Protocol	Port range	Source	Allow/Deny
99	All traffic	All	All	0.0.0.0/0	Deny
101	All traffic	All	All	0.0.0.0/0	Allow
*	All traffic	All	All	0.0.0.0/0	Deny

Edit inbound rules

<

1

>

Similarly for outbound rules as well.

We can also add a specific Ip address, for which it can **deny inbound/outbound access**. But this is not possible in Security group.

The number of instances (IPv4) you can create in a subnet determined by the **CIDR block address -5 reserved ip address**.

(Minimum /28 and Maximum /16)

For eg. A common /24 subnet can have 251 instances.

$$32 - 24 = 8$$

$$2^8 = 256$$

$$256 - 5 \text{ reserved ip address} = 251 \text{ instances.}$$